

Propeller Ground and Ceiling Effect in Forward Flight

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ABSTRACT

The study of the fixed-pitch small-scale propeller in ground/ceiling effect is extended to forward flight conditions. Force-based experiment and smoke flow visualization is conducted at the University of Dayton Low Speed Wind Tunnel. The propeller is mounted at different incidence angles to simulate the transition to forward flight from hover and forward flight conditions. The effect of propeller incidence angle seems to be negligible at $J=0$. The power required at constant thrust shows an increasing trend after a threshold advance ratio for each h/D and then decreases steeply upon further increase in advance ratio. Smoke flow visualization revealed the presence of three distinct zones separated by the onset of the fountain vortex and the onset of the ground vortex at each ground distance. At advance ratios below the onset of the fountain vortex lies the optimal ground effect zone. An increase in advance ratio beyond this threshold will result in power increment due to the fountain vortex. Further increase in advance ratio initiates the onset of the ground vortex where the ground effect is significantly reduced. The onset of the fountain vortex and the ground vortex occurs at lower advance ratios for a propeller at an incident angle to the ground.

NOTATION

C_T	=	Propeller thrust coefficient; $C_T = \frac{T}{\rho n^2 D^4}$
C_P	=	Power coefficient; $C_P = 2\pi C_Q$
C_Q	=	Propeller torque coefficient; $C_Q = \frac{Q}{\rho n^3 D^5}$
D	=	Propeller diameter, (m)
h	=	Distance to the ground-plate, (m)
J	=	Advance Ratio; $J = \frac{V_\infty}{nD}$
n	=	Propeller rotational speed per second,
P	=	Power, (W)
Q	=	Torque, (Nm)
R	=	Propeller radius, (m)
T	=	Thrust, (N)
V_∞	=	Freestream Velocity, (m/s)
ρ	=	Air density, (kg/m^3)
θ	=	Propeller incidence angle, (deg)
γ	=	Propeller pitch, (m)
σ	=	Propeller solidity; $\sigma = \frac{s}{0.25\pi D^2}$

Subscripts:	
ICE	= In Ceiling Effect
IGE	= In Ground Effect
OCE	= Out of Ceiling Effect
OGE	= Out of Ground Effect

INTRODUCTION

Ground effect on lifting rotor/propeller performance during hovering has been studied theoretically (Ref.1) and experimentally (Ref.2 – Ref.6) for decades. Placing the propulsor adjacent and parallel to a ground-plate offers two benefits: an increase in thrust-produced, and a decrease in power-required, for a given rotational speed, while the latter benefit is usually ignored in many small-scale propeller-driven UAV studies. Multiple ground effect models for hovering have been developed for large rotors (Ref.1, Ref.2, Ref.3) with power loading CT/σ one order of magnitude lower than that of propeller rotors use on small UAVs. However, most of the models do not produce a good prediction on small scale propellers widely used by smaller scale UAVs, which is discussed in the prior study (Ref.7 Ref.8) where multiple APC thin electric propellers were tested where normalized power required at constant thrust, P_{IGE}/P_{OGE} , was used to evaluate the performance of the propeller in the ground and ceiling effect. An algebraic prediction model was developed and validated using multiple APC propellers that is able to predict the performance within 6% error.

Rotor ground effect in forward flight has also been widely studied for larger rotors experimentally (Ref.2, Ref.3, Ref.6). Ref.2 conducted experiments on a full-scale Sikorsky S-51 and the investigations in Ref.3 and Ref.6 were conducted in a

wind tunnel on a large rotor. Ref.3 specifically studied the rotor ground effect at a small advance ratio ($J < 0.08$ range) while the other two references studied a wider advance ratio range up to $J = 0.7$. The results from these references are shown in Figure 1. The IGE power required at constant thrust is normalized by the OGE condition at $J = 0$. An increment in P_{IGE}/P_{OGE} with the increment of J at $J < 0.1$ is found in all literature at $h/D < 0.5$, indicating a loss in rotor performance at small forward flight speed in ground effect. This phenomenon is caused by the rotor leading edge recirculation/vortical flow region (Ref.6 Ref.9) at a small range of advance ratios. Moreover, none of the existing models generated from experimental data or computational fluid dynamics analysis, could provide a good prediction of such phenomenon (Ref.9). With a further increment of J , the ground effect benefits vanish, as the P_{IGE}/P_{OGE} collapse for both IGE and OGE performance at $J > 0.25$. Our prior work (Ref 7 and Ref 8) covered the ground and ceiling effect of small-scale propeller in hovering conditions. The current investigations expand our prior study to ground and ceiling effect in forward flight, the latter one has never been investigated since large-scale rotorcraft is very unlikely to be operated in a confined space close to any ceiling. It is more likely for a propeller-driven UAV to be operated in such condition. Hence, the present study will fulfill the aforementioned knowledge gap.

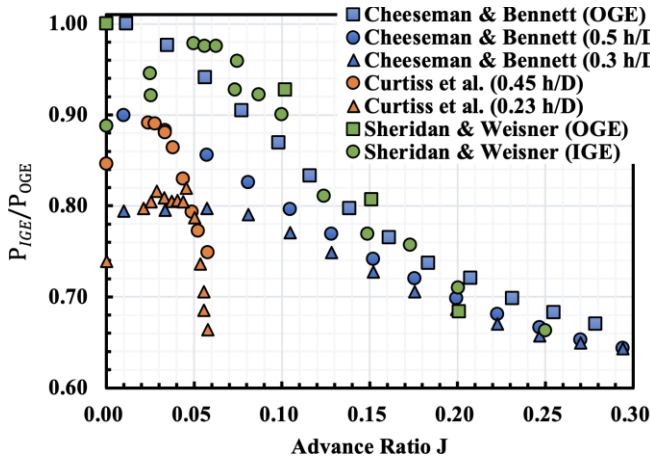


Figure 1. Normalized power required at constant thrust v.s advance ratio from literature (Ref. 2 Ref. 5 Ref. 6)

EXPERIMENTAL SETUP

Force Based Experiment

Experiments were conducted in the University of Dayton Low Speed Wind Tunnel (UD-LSWT) in the open jet configuration. The UD-LSWT has a 16:1 contraction ratio, 6 anti-turbulence screens, and 0.762 m by 0.762 m open jet test sections. A 0.914 m by 1.829 m aluminum alloy sheet with 0.005 m thickness was used to simulate the ground plate for the experiment. All propellers were driven by an E-Flite Power 60-400kV brushless out-runner electric motor, controlled by a Hobbywing Platinum V3 ESC under governor mode to maintain constant rotation speed. The propeller and motor are driven by a PSW 30-108 constant-voltage power

supply. A photodiode was used to measure propeller RPM. The schematic of the setup is given in Figure 2.

The propeller motor was attached to an ATI Industrial Automation Mini-40 six-component force balance. Thrust is the axial-force along the z-axis of the sensor, and torque is a rotational moment about the z-axis. The balance bolts to an aluminum-extrusion frame secured to the inlet of the plenum. The balance has a maximum of 120N force on its Z-axis and maximum torque of 2 Nm about its Z-axis. The encountered thrust and torque range was 5N to 25N and 0.1 Nm to 0.35 Nm, respectively. The average uncertainty in the force measurement after filtration is 7.2%, while the average uncertainty in torque is 6.1%. However, the results presented show excellent repeatability.

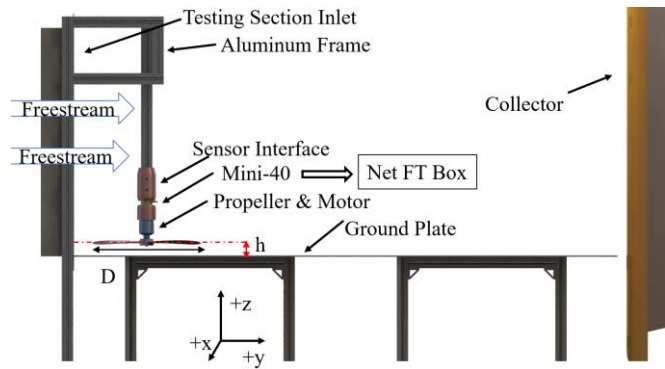


Figure 2. Schematic of Experimental Setup of the Propeller-Ground Test Stand in UD-LSWT

Table 1 shows the testing matrix of the experiment. The propeller is mounted at a 90-degree and 75-degree incidence angle (θ) for the current investigation. The propeller ground distance is measured from the center of the propeller hub to the ground plate. This distance is then normalized by the propeller diameter resulting in the normalized ground distance, h/D . Due to the propeller tip interference at 75-degree incidence angle, the minimum ground distance h/D is greater than the 90-degree incidence angle. To achieve different advance ratios, the propeller rotational speed is kept constant, while the wind tunnel freestream velocity is changed accordingly, between 0 and 15 m/s.

Table 1. Testing Matrix

Testing Condition	Value
Propeller	APC 14x6 11x7
Diameter	0.3556 0.2794 m
Number of Blades	2
Solidity	0.0808 0.0849
Rotational Speed	83.3 100 116.7 RPS
Freestream Velocity	0-15 m/s
Normalized Ground Distance	0.07-0.86
Propeller Incidence Angle	75 90 Degree

To investigate the ceiling effect, the propeller orientation and rotational direction is reversed. The aluminum alloy sheet then serves as a ceiling plate. For the forward flight

configuration at $\theta = 75^\circ$, the propeller tilting angle with respect to the freestream is also reversed, so that the direction of the propeller wake is away from the ceiling plane.

Phase-lock Flow Visualization

Smoke based flow visualization is performed using the phase-lock technique to further understand the propeller flow field in ground effect forward flight. The Concept ViCount 5000 smoke machine, pressurized by nitrogen at 100 Kpa is used to generate the oil-based smoke. The smoke machine is connected to a smoke wand with a diameter of 6mm. Two 300 mW, 532 nm laser aligned with a 30-degree fan angle Powell lens are used to illuminate the smoke in the test section. The Imperx 2021 CCD camera is used to take the flow visualization image, with a Canon EF 50mm f/1.4 lens. A 50 mW, 532 nm laser, and a photodiode are used to trigger the Quantum Composers 9600 pulse generator, which is then used to trigger the camera shutter. By changing the delay on the QC pulse generator, the flow passing different propeller phases can be captured.

RESULTS

A. Effect of Propeller Incidence in Ground Effect in Hovering

As the propeller incidence angle changes in forward flight, it is important to investigate the effect of propeller incidence angle on the ground effect in hovering ($J = 0$). Figure 3 shows the IGE thrust coefficient, normalized by the OGE thrust coefficient for both propellers at different incidence angles. Averaged result from different propeller rotational speeds is shown here, as the effect of Reynolds number is found to be less than 5%. Positive h/D represents ground effect while negative h/D represents ceiling effect.

For all cases, a rise in C_{TIGE}/C_{TOGE} approaching $h/D = 0$ is nearly symmetric. The 14x6 propeller, which has a lower γ/D ratio shows a greater thrust augmentation in ground effect, which agrees with the result in our previous study (Ref.7 Ref.8). When tilting the propeller forward at a lower incidence angle, slight differences in thrust is observed in both propellers in ground effect at the same h/D value, while the difference in ceiling effect is almost negligible.

Figure 4 shows the normalized power coefficient for the cases shown in Figure 3. A decrement in C_p is found when h/D approaching to zero for both ground and ceiling effect. Again, the phenomenon is also found in our previous study, with a lower γ/D ratio propeller result in a higher power saving in ground effect. Similar to the thrust coefficient, a small increase in power coefficient is found in all ground effect cases, while no significant differences are found in the ceiling effect.

A common representation of the combined effect of propeller thrust and torque performance in ground effect is the “normalized power” (P_{IGE}/P_{OGE}), at constant thrust. To

calculate this, IGE and OGE thrust measured at all propeller rotational speeds tested is used. A reference value of rotational speed is fixed, at which OGE thrust is noted. For every h/D of interest, interpolation of the measured thrust at that h/D across the rotational speed values is done, to find that particular rotational speed, for which the thrust equals the aforementioned OGE thrust. Then, at this particular rotational speed, the IGE power is obtained from interpolating across the power-measurement data set, at that h/D . Results are averaged across OGE thrust levels – that is, across propeller rotational speed values, which is shown in Figure 5 below, along with the experimental result from our previous study.

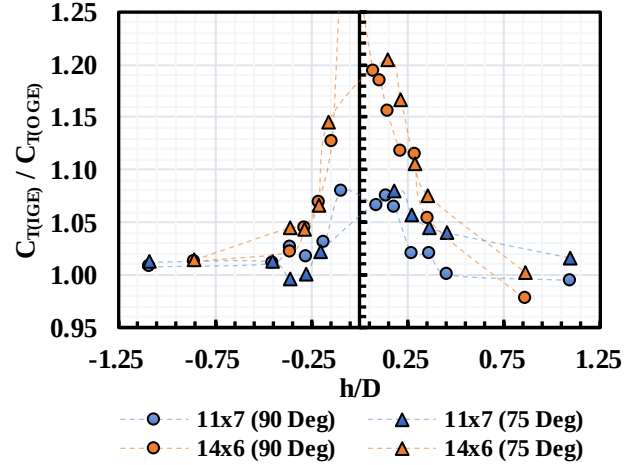


Figure 3. Normalized thrust coefficient for different incidence angles

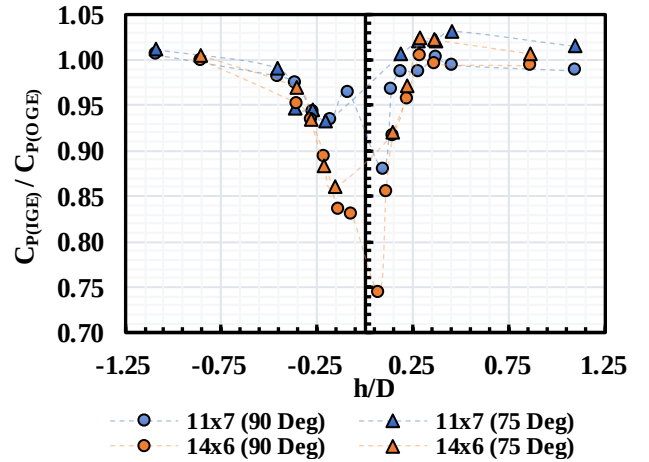


Figure 4. Normalized power coefficient at different incidence angles

As a result of thrust augmentation and power reduction in the ground and ceiling effect, P_{IGE}/P_{OGE} reduces with approaching $h/D = 0$, with the result at $h/D > 0$ almost identical to the result at $h/D < 0$. Due to the small increment in thrust and power coefficient at $\theta = 75^\circ$ at $h/D > 0$ seen in Figure 3 and Figure 4, the resulting P_{IGE}/P_{OGE} overlaps on the curve of $\theta = 90^\circ$ for both propellers, as well as the

experimental result from the author's previous study. Hence, the effect of propeller incidence angle is found to be negligible in ground and ceiling effect under hovering conditions at $75^\circ < \theta < 90^\circ$.

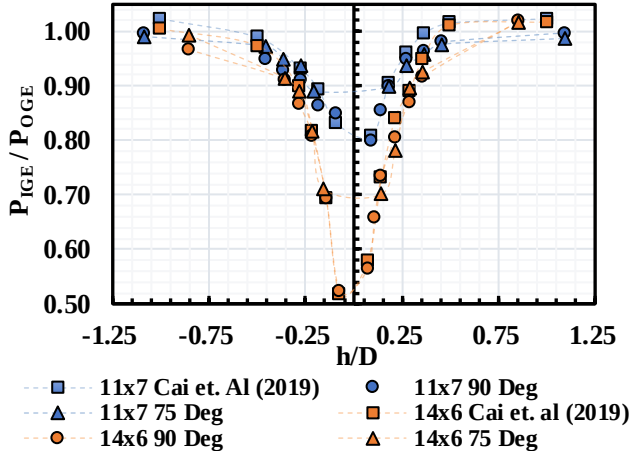


Figure 5. Normalized power required at constant thrust at different incidence angles

B. Propeller Ground Effect in Forward flight

An average of 5.5% difference in the thrust coefficient and 4.8% difference in the power coefficient is found between the lowest rotational speed and highest rotational speed at the same advance ratio indicating negligible Reynolds number effects. Thus, the experimental result from the 100 RPS case was chosen to be a good representative to show the overall propeller performance in ground and ceiling effect in forward flight.

Figure 6 shows the normalized thrust coefficient, C_{TIGE}/C_{TOGE} v.s advance ratio, J , at different normalized ground distances, h/D for the APC 14x6 propeller at $\theta = 90^\circ$, which represents the different stages of transition to forward flight. In Figure 6, C_T of each datapoint is normalized by $C_{TOGEJ=0}$, which is C_T at OGE and $J = 0$, which is the propeller OGE performance in hovering. An increment in C_T is found at $J > 0.2$ with the increment of J . The thrust augmentation is purely due to the increment in freestream since the propeller RPS was held constant. The highest h/D case tested, $h/D = 0.86$ shows similar trend as the OGE condition where the C_T remains constant in $0 > J > 0.2$ and increases when $J > 0.2$. Indicating a minor ground effect on the propeller performance at such h/D . With increase in advance ratio.

At smaller h/D , the effect of ground becomes prominent, as an increment in C_T is found at $h/D < 0.36$ when compared to OGE. However, at $h/D < 0.29$, the change in C_T vs J does not follow the same trend as seen in OGE or higher h/D cases. While C_T remains constant at $J < 0.1$, a decrement in C_T is found between $0.2 > J > 0.3$, except for the extreme ground effect cases at $h/D = 0.07$. Eventually, the value of C_T

overlaps at $J = 0.3$, indicating a reduction of ground effect benefit of thrust augmentation, though there is still a 10% increment in thrust when compared to the OGE case. At $J > 0.3$, all cases show a similar trend of increase in C_T , with a clear difference between each h/D case, where a lower h/D results in higher thrust augmentation. It is worth noticing that further investigation is required to understand the cause of the increment in C_T at higher J range.

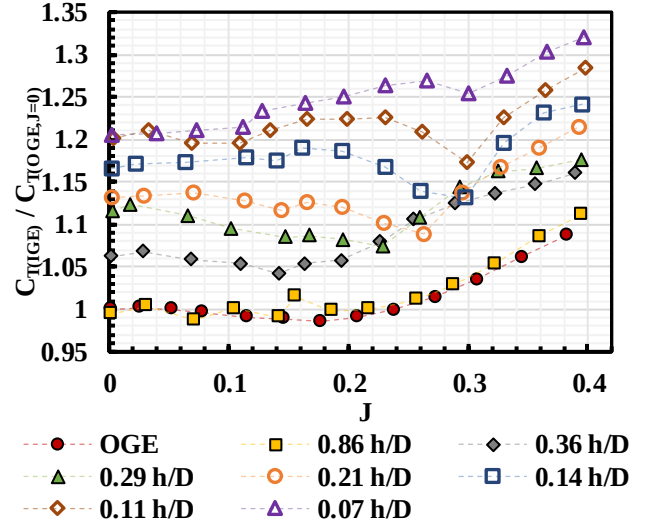


Figure 6. Normalized thrust coefficient ($C_{TIGE}/C_{TOGEJ=0}$) at $\theta = 90^\circ$ and 100 RPS in ground effect

To eliminate the effect of freestream, an alternative normalization method is to normalize C_T of each h/D case by the C_T at OGE condition under the same J . This is shown in Figure 7. At large h/D , the overall trend is similar to Figure 6. However, at $h/D \leq 0.14$, a clear increment in C_{TIGE}/C_{TOGE} is observed at $0.1 < J < 0.2$, followed by a clear decrement at $0.2 < J < 0.3$ which is discussed previously. Interestingly, at $J \geq 0.4$, the value of C_{TIGE}/C_{TOGE} is similar to $J = 0$, indicating the effect of freestream in overall ground effect is minimized, which is also seen in the case of 83.3 RPS case, where the equivalent J extended to 0.5.

Since the data normalization with OGE at different J values bring more insight into the effect of ground on the propeller performance, the normalization in the following sections will be performed with the same method. An in-depth discussion of the changes in OGE propeller performance due to propeller incidence angle is included in our previous study (Ref 11).

Figure 8 shows C_{TIGE}/C_{TOGE} when tilting the propeller forward at $\theta = 75^\circ$ at different advance ratios, which represent the propeller performance in forward flight. Similar to the overall trend in Figure 7, changes in C_T is minor at $J < 0.1$. A small increment in C_T is again observed between $0.1 < J < 0.15$ at $h/D \leq 0.14$. However, the decrement in C_T occurs at lower J value when compared to $\theta = 90^\circ$, between $0.15 < J < 0.25$. At $J > 0.25$, the overall trend again resembles Figure 7. Again, the difference of C_{TIGE}/C_{TOGE}

between $J = 0$ and $J = 0.4$ is negligible for $h/D > 0.21$, while a 4% loss is found for smaller h/D cases.

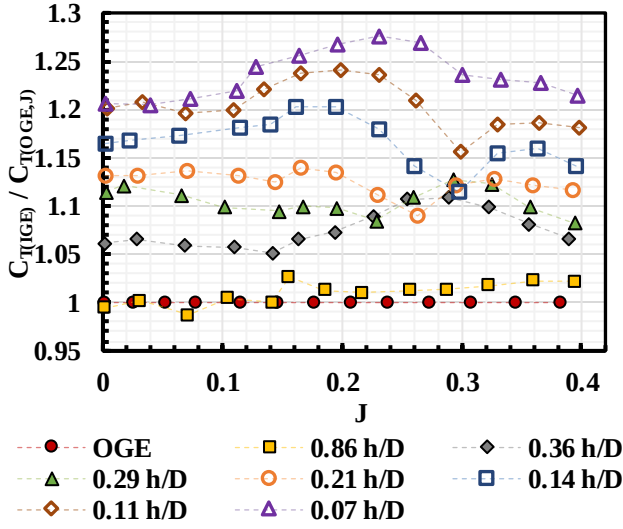


Figure 7. Normalized thrust coefficient at $\theta = 90^\circ$ and 100RPS in ground effect

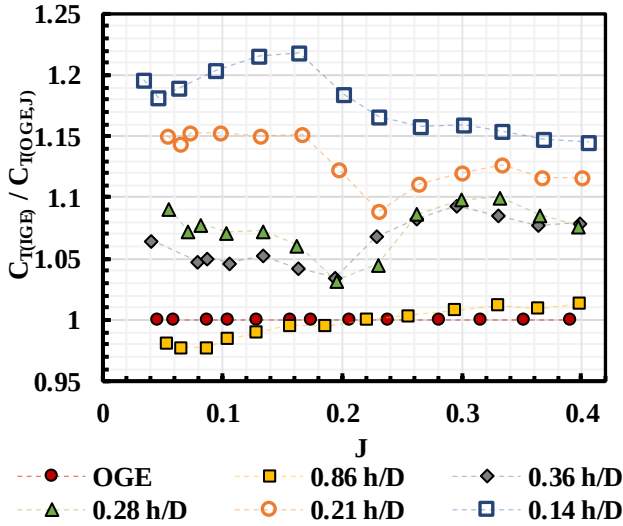


Figure 8. Normalized thrust coefficient at $\theta = 75^\circ$ and 100RPS in ground effect

Similar to C_T , C_p also shows significant departures from OGE due to ground effect and advance ratio, as seen in Figure 9. Recall the result in Figure 4, the ground effect benefit in hovering is noticeable on C_p only at $h/D < 0.28$, this benefit reduces with the increment of J . It is worth noticing that for all cases at $h/D < 0.28$, C_{pIGE}/C_{pOGE} exceeds 1.0 at some point between $0.2 < J < 0.4$, with lower h/D cases extends its IGE C_p benefit to higher J . At $J \sim 0.4$, the CP curves for all h/D cases collapse onto each other indicating the loss in ground effect benefits.

Figure 10 shows the normalized power coefficient for $\theta = 75^\circ$. Again, the overall trend resembles Figure 9. However,

for cases at $h/D < 0.28$, C_{pIGE}/C_{pOGE} no longer exceeds 1.0 at higher J , while the curve for different h/D cases collapse at $J \sim 0.3$ which is lower than $\theta = 90^\circ$. Recall that for C_T at $\theta = 75^\circ$, the changes in C_T also occur at a lower J value, indicating tilting the propeller forward ‘advances’ the changes in propeller performance with respect to J .

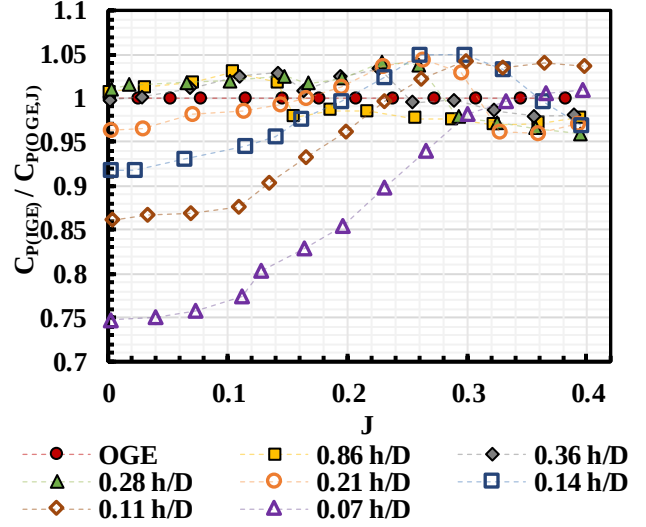


Figure 9. Normalized power coefficient at $\theta = 90^\circ$ and 100RPS in ground effect

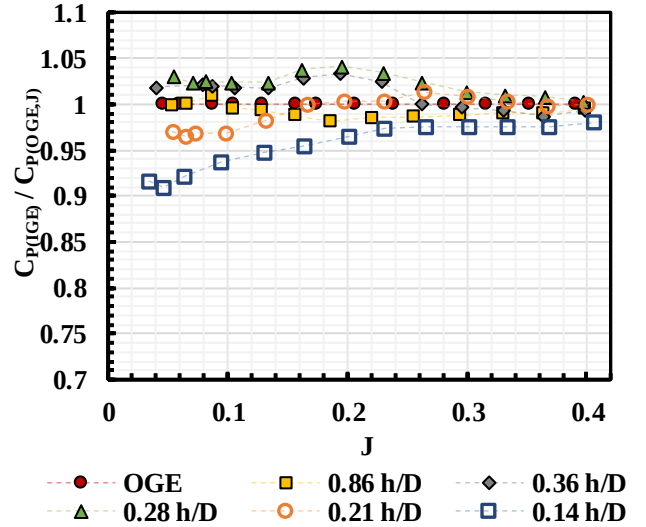


Figure 10. Normalized power coefficient at $\theta = 75^\circ$ and 100RPS in ground effect

The calculation of the normalized power required at constant thrust, P_{IGE}/P_{OGE} in forward flight is slightly different than the hovering condition. The data from multiple propeller rotational speeds at the same freestream velocity is used. At every h/D of interest, under each freestream velocity, an interpolation of the thrust and power v.s the propeller rotational speed is created based on the measured thrust and power at this specific freestream velocity. Then, a thrust level of interest is selected. Utilizing the aforementioned interpolation, a particular rotor rotational speed is found at the

selected thrust level. Then, at this particular rotational speed, the corresponding power required is obtained from the other interpolation. With the freestream velocity and propeller rotational speed known, the advance ratio is then calculated. This process is repeated at each freestream velocity to produce the result of P_{IGE} vs J for one h/D case and the entire process is repeated at each h/D . The IGE power is then normalized by the power of OGE at the same thrust level and advance ratio to produce P_{IGE}/P_{OGE} . A thrust level of 15N is selected in this section, which is the medium thrust level for the APC 14x6 propeller.

The normalized power at constant thrust for $\theta = 90^\circ$ is shown in Figure 11. For large h/D , the result again resembles the OGE trend. For smaller h/D cases at $J < 0.15$, a small increment in P_{IGE}/P_{OGE} is observed for all ground distance cases, which is a result of the increment of C_p seen in Figure 9 at the same J range. With the increment of freestream velocity, between $0.15 < J < 0.35$, a clear increment of P_{IGE}/P_{OGE} is found for all h/D cases with the increment of J , until collapse with other curves. After the data collapse, curves for each h/D case deviates with a lower P_{IGE}/P_{OGE} value for smaller h/D cases. An overall difference 12% in P_{IGE}/P_{OGE} between $J = 0$ and 0.4 is found for extreme ground effect cases at $h/D < 0.14$, due to the loss of ground effect benefit in power reduction shown in Figure 9, as expected. However, at higher h/D cases, the difference in P_{IGE}/P_{OGE} between $J = 0$ and 0.4 seems negligible, as a result of ‘thrust recovery’ at high J seen in Figure 7.

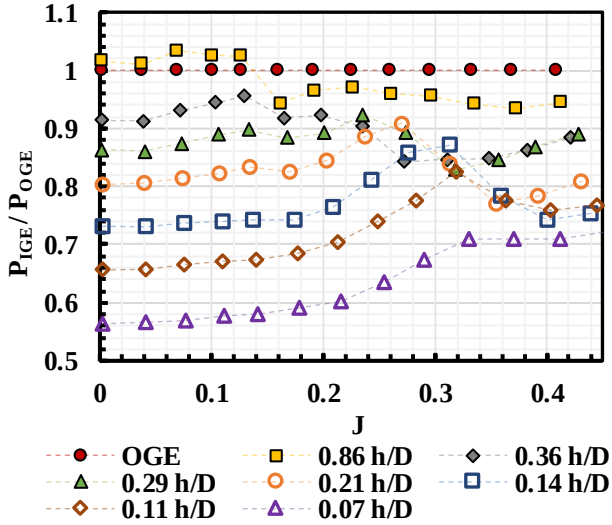


Figure 11. Normalized power required at constant thrust $\theta = 90^\circ$ and 100RPS in ground effect

P_{IGE}/P_{OGE} for $\theta = 75^\circ$ is shown in Figure 12, with a very similar trend seen in Figure 11. As discussed previously, tilting the propeller forward ‘advances’ the changes in propeller performance with respect to J due to the blade tip being much closer to the ground when compared to the hub. The increment in P_{IGE}/P_{OGE} occurs at $J = 0.1$ while the collapse of curves occur at $J = 0.25$, both are at lower J when

compared to the result for $\theta = 90^\circ$. Again, the difference in P_{IGE}/P_{OGE} between $J = 0$ and 0.4 seems negligible for $h/D > 0.14$, while a 10% difference is found at smaller h/D cases.

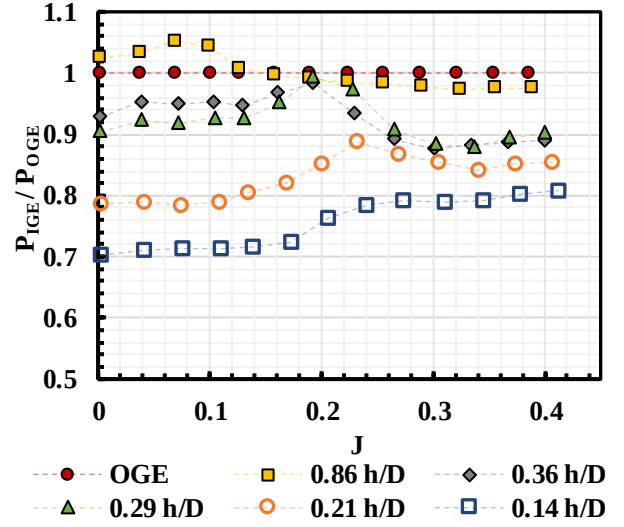


Figure 12. Normalized power required at constant thrust $\theta = 75^\circ$ and 100RPS in ground effect

C. Propeller Ceiling Effect in Forward flight

The changes in propeller performance in ceiling effect forward flight is then investigated. For all ceiling effect cases, the normalization is done with respect to the OCE cases where the propeller orientation and rotational direction are reversed when compared to the OGE cases, to eliminate any installation interference of the aluminum frame.

Figure 13 and Figure 14 show the normalized thrust coefficient, C_{TICE}/C_{TOCE} at $\theta = 90^\circ$ and 75° . The overall trend for both incidence angles is very similar. At $J < 0.1$, a clear reduction in C_T is observed with the increment of J for all cases, the trend is then reversed between $0.1 < J < 0.3$, which differs significantly from the IGE cases in Figure 7. With a further increment in J , C_{TICE} reduces slightly, except for the extreme ceiling effect case at $h/D = -0.07$. Interestingly, the difference of C_T between $J = 0$ and 0.4 is again very small, which is also observed in the ground effect cases.

Normalized power coefficient for both $\theta = 90^\circ$ and 75° are shown in Figure 15 and Figure 16. For $\theta = 90^\circ$ at $h/D < -0.14$, C_{PICE}/C_{POCE} remain relatively constant with the increment in J , and merge with other curves between $0.2 < J < 0.4$. After the curve merges, C_p increase with further increment of J and then reach 1.0 at $J = 0.4$, resulting in no ceiling effect power benefit. On the other hand, the overall trend for $\theta = 75^\circ$ resembles the $\theta = 90^\circ$ case. However, it is worth noticing that the increment in C_p occurs at a much lower J of 0.1, instead of $J \sim 0.25$.

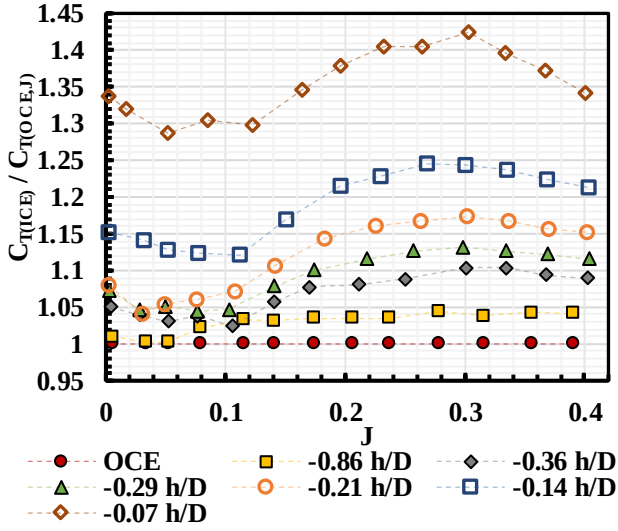


Figure 13. Normalized thrust coefficient at $\theta = 90^\circ$ and 100RPS in ceiling effect

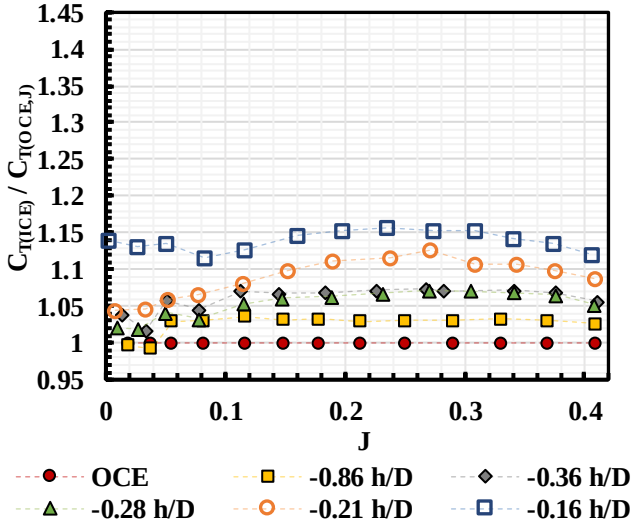


Figure 14. Normalized thrust coefficient at $\theta = 75^\circ$ and 100RPS in ceiling effect

An abnormal decrease of C_p is observed for the extreme ceiling effect cases at $h/D = -0.07$ between $0 < J < 0.12$. This is very likely a result of blade stall recovery as the distance between the propeller and the ceiling plane is small, at low J , not enough propeller upstream inflow is provided, resulting in an increment in blade effect angle of attack, causing a propeller blade stall. With the increment in J , the inflow of the propeller increases, which reduces the effect rotor angle of attack, reducing the local blade drag, which reduces the power consumption of the propeller.

To combine the effect of thrust argumentation and power reduction, the normalized power required at constant thrust, P_{ICE}/P_{OCE} is calculated. The result for $\theta = 90^\circ$ is shown in Figure 15. Except for $h/D = -0.07$ case, where P_{IGE}/P_{OCE} remains constant until $J = 0.3$, all cases have a small increment in P_{ICE}/P_{OCE} at $J < 0.15$, the trend is then reversed

between $0.15 < J < 0.3$. Since the changes in C_p within the J range is small, the changes in P_{ICE}/P_{OCE} is mainly a result of the changes in C_T seen in Figure 13 the same J range. At higher J range, the value of P_{ICE}/P_{OCE} increases for all cases due to the loss in IGE power savings seen in Figure 15.

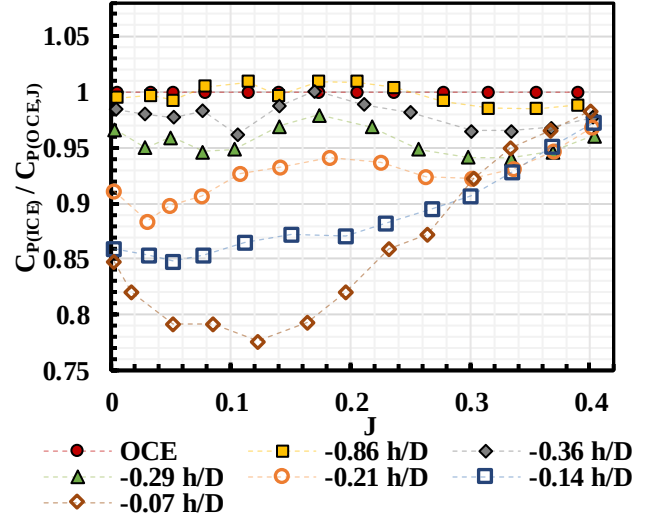


Figure 15. Normalized power coefficient at $\theta = 90^\circ$ and 100RPS in ceiling effect

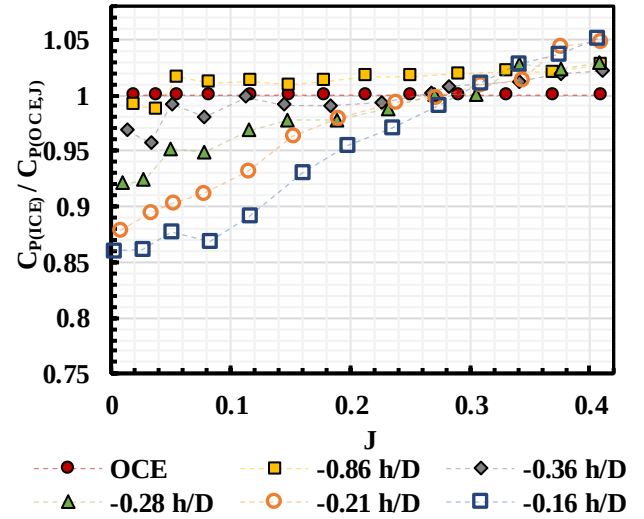


Figure 16. Normalized power coefficient at $\theta = 75^\circ$ and 100RPS in ceiling effect

When tilting the propeller forward, the increment in P_{ICE}/P_{OCE} at $J < 0.15$ preserves for all h/D tested. However, the value of P_{ICE}/P_{OCE} does not reduce between $0.15 < J < 0.3$, instead, it remains relatively constant at this J range. The increment in P_{ICE}/P_{OCE} occurs again at $J > 0.3$ which is also seen in Figure 17. Differ from the ground effect, a clear reduction of P_{ICE}/P_{OCE} between $J = 0$ and 0.4 is observed, indicating a significant effect of freestream on the propeller performance in the ceiling effect.

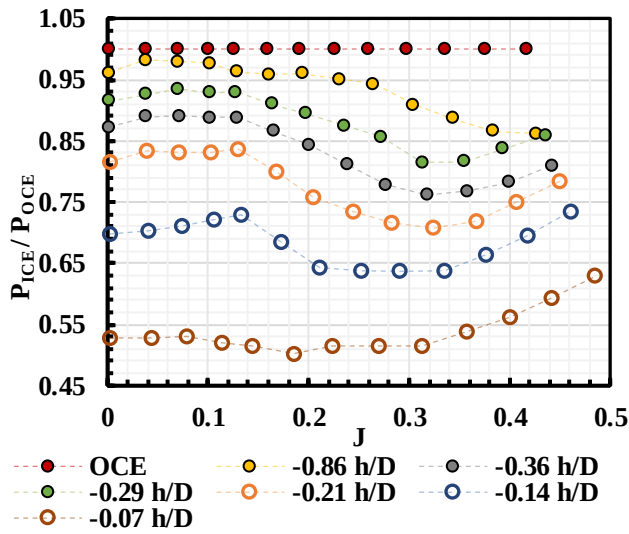


Figure 17. Normalized power required at constant thrust $\theta = 90^\circ$ and 100RPS in ceiling effect

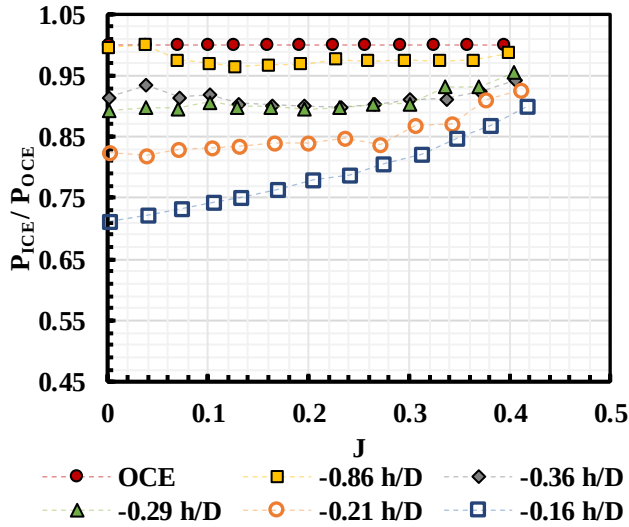


Figure 18. Normalized power required at constant thrust $\theta = 75^\circ$ and 100RPS in ceiling effect

D. Ground Effect Phase-lock Smoke Flow Visualization

Smoke based flow visualization was performed on the APC 14x6 propeller to better understand the flowfield around the propeller in ground effect. The discussion in this section mainly focuses on the flowfield near the leading edge of the propeller disk. Side view of the propeller is presented with the freestream coming from the left to right of the image. Figure 19 shows the flow visualization at $J = 0$ and $J = 0.1$ for incidence angle $\theta = 90^\circ$ and $\theta = 75^\circ$ at $h/D = 0.2$. At $J = 0$, the tip vortices spread radially in all directions due to ground effect. With an increase in J to 0.1, the outward propeller momentum is met with the freestream momentum however, the propeller wake momentum remains dominant.

When further increasing the advance ratio to 0.2, the tip vortex is still being pushed outward radially for the $\theta = 90^\circ$

case. A ground fountain vortex is formed at the propeller tip for $\theta = 75^\circ$, while the same phenomena occur for $\theta = 90^\circ$ case at $J = 0.3$, as seen in Figure 21. Recall from the C_T result in Figure 7 and Figure 8, at decrement in C_T is found between $0.15 < J < 0.25$ for $\theta = 75^\circ$, which correlates to the formation of this ground fountain vortex, which is the recirculation region formed by the propeller wake being injected back into the propeller disk (Ref 12., Ref 13).

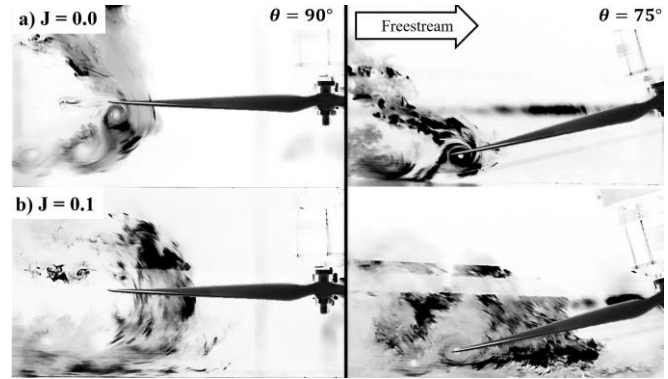


Figure 19. Flow visualization for $\theta = 90^\circ$ (left) and $\theta = 75^\circ$ (right) at a) $J = 0$ and b) $J = 0.1$ and $h/D = 0.2$

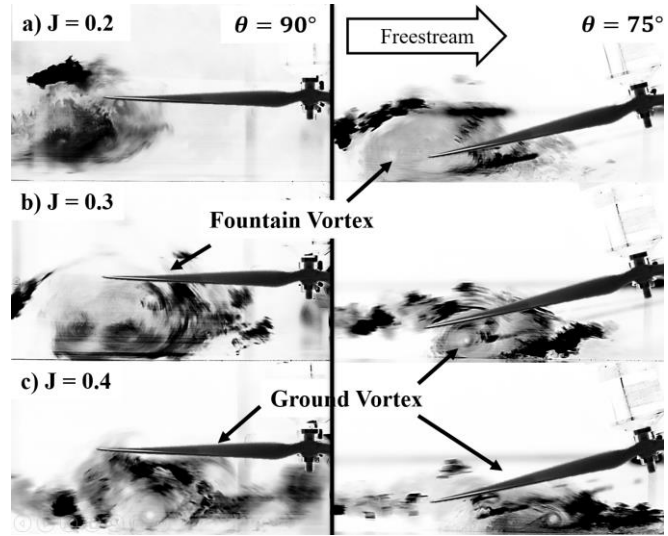


Figure 20. Flow visualization for $\theta = 90^\circ$ (left) and $\theta = 75^\circ$ (right) at a) $J = 0.2$, b) $J = 0.3$ and c) $J = 0.4$ and $h/D = 0.2$

However, with the increment in advance ratio, another type of recirculation region is formed in between the propeller disk and the ground called the ground vortex, which is observed at the $\theta = 75^\circ$ case at $J = 0.3$. Recall that the C_T and C_P difference between each h/D case is relatively small at this advance ratio range (Figure 7 - Figure 10), indicating a reduction in the effect of the ground distance on propeller performance. Similarly, this ground vortex is formed at $0.3 < J < 0.4$ for the $\theta = 90^\circ$ case. For $\theta = 75^\circ$ at the highest advance ratio investigated, the location of the ground vortex moved further downstream when compared to $J = 0.3$. Apparently, the same flow pattern for $\theta = 90^\circ$ occurs at

lower J range for the $\theta = 75^\circ$, as a result of the effect of pitching the propeller forward.

Figure 21 emphasizes the effect of ground distance on the propeller flowfield in ground effect. At $J = 0.3$, the fountain vortex occurs at $h/D = 0.1$ and 0.2 . At higher h/D , the formation of the ground vortex between the propeller disk and the ground plate replaced the fountain vortex. At $J = 0.4$, the ground vortex occurs at the tip of the propeller disk at $h/D = 0.1$. The location of the ground vortex moves downstream at $\sim 75\%$ disk radius location for $h/D = 0.2$. However, at $h/D = 0.3$, the ground vortex disappears, where the IGE benefit also reduces significantly as seen in Figure 11. It is also worth noting that the changes in ground vortex location due to the increment of h/D is similar to the effect of the increment of J for $\theta = 75^\circ$ when h/D is held constant, as seen in Figure 20. In other words, the effect of increasing h/D has some similarities when compared to the effect of increasing J .

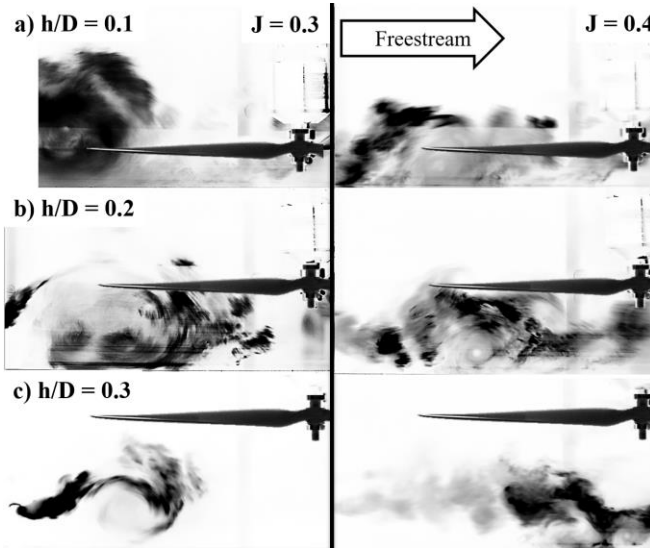


Figure 21. Flow visualization for $\theta = 90^\circ$ at $J = 0.3$ (left) and $J = 0.4$ (right) at a) $h/D = 0.2$, b) $h/D = 0.3$ and c) $h/D = 0.3$.

Figure 22 separates the propeller performance in ground effect in forward flight into three different regions, based on the flow visualization result discussed in this section. At a small J range, where the propeller wake dominates, the ground effect benefit is at its optimal state, which resembles the hovering condition. With the increment in J , initiates the formation of the fountain vortex which disrupts the thrust produced and increases the power required. The strength of the fountain vortex continues to grow with the increment in advance ratio. With a further increment of J , a ground vortex formed in between the propeller disk and the ground where the ground effect benefit is significantly reduced.

Figure 23 summarizes the different flow regions shown in Figure 22 for both $\theta = 90^\circ$ and $\theta = 75^\circ$, in the format of J v.s h/D . At $h/D > 0.5$, the IGE benefit is negligible. At a

higher h/D , the onset of fountain vortex formation and the ground vortex formation occurs a lower advance ratio. However, decreasing the h/D delays the onset of these vortices to a comparatively higher advance ratios. When the inclination angle is reduced, the onset of fountain vortex and the ground vortex occurs at a significantly lower advance ratios as compared to $\theta = 90^\circ$.

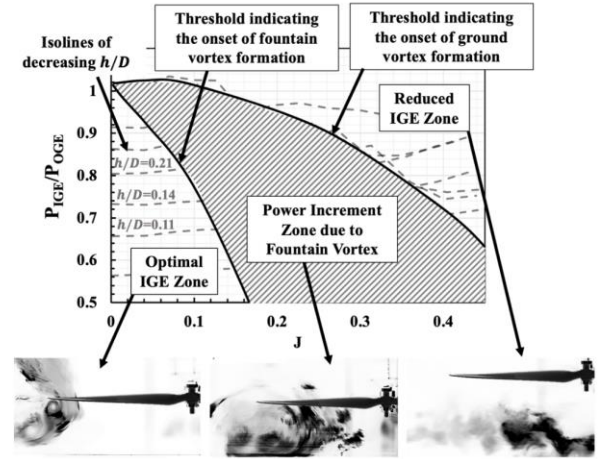


Figure 22. Zones of propeller performance marked by three distinct flow fields

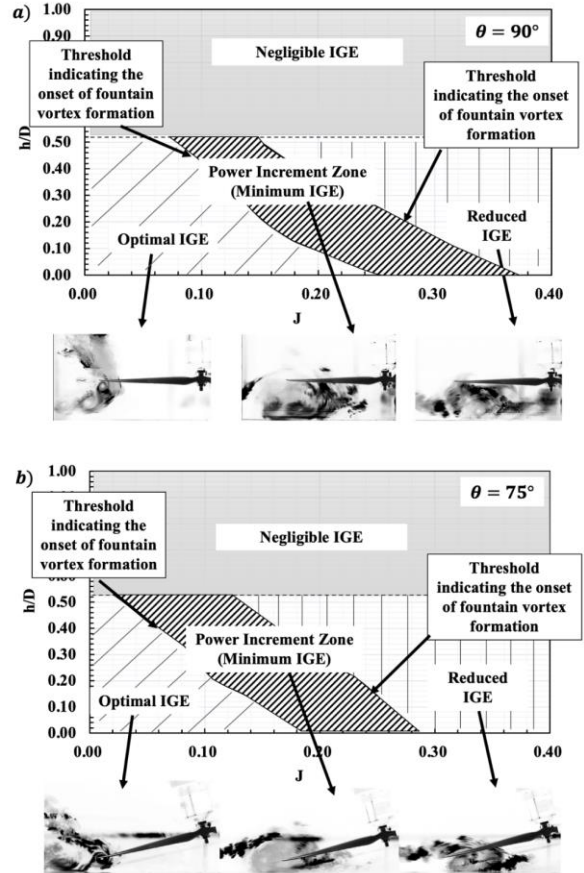


Figure 23. Propeller IGE performance region for a) $\theta = 90^\circ$ and b) $\theta = 75^\circ$, plotting with J v.s h/D

CONCLUSIONS

The changes in propeller performance at different incidence angle in hover and in forward flight was studied with and without the ground and ceiling effect. The ground effect performance in hover was found to be insensitive to the propeller inclination angle in the range studied, resulting in a similar performance in thrust and power between 90° and 75° degree orientations.

The effect of freestream on the propeller performance was negligible at $J < 0.1$ for all h/D where the propeller momentum is dominant. With the increment of freestream velocity, at $h/D < 0.3$, a ground fountain vortex is formed, resulting in a reduction of propeller thrust generated and increment in power required, thus, a reduction in propeller performance. Further increment of advance ratio results in the formation of a steady ground vortex between propeller disk and ground plate, where the ground effect benefits in both thrust and power are minimized. The location of this ground vortex moves further downstream with the increment of J . At $h/D < 0.21$, the ground vortex perseveres till $J = 0.4$, resulting in a further increment in thrust generation. At higher h/D , this ground vortex vanished, resulting in a performance similar to the OGE condition.

For the ceiling effect, the changes in propeller thrust generation differ from the ground effect. A loss in thrust generation is observed at a small advance ratio with the increment of advance ratio, the trend is reversed between $0.1 < J < 0.3$. At a higher advance ratio, a small decrement in thrust is then observed. On the other hand, the power consumption resembles more or less the overall trend seen in ground effect, with a minor change at small advance ratio, except for the extreme ceiling effect cases. C_p then increases with further increment in advance ratio and reaches OCE level at $J \sim 0.3$, which agrees with the ground effect performance.

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